Status	Finished
Started	Friday, 30 October 2026, 12:08 AM
Completed	Friday, 30 October 2026, 12:12 AM
Duration	4 mins 21 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists
b. It does not work with negative numbers
c. It requires the list to be sorted ✓
d. It is slower than linear search for small lists

Question **2**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Binary ✓
b. Hash
c. Linear
d. Both (1) & (3)

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Hashing
b. Linear search ✓
c. Dynamic search
d. Binary search

Question **4**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n-1
b. n/2
c. log n
d. n ✓

Question **5**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Sequential search
b. Hash search
c. Binary search ✓
d. Linear search

Question **6**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Linear search ✓
b. Interpolation search
c. Hash search
d. Binary search

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There must be a mechanism to delete and/or insert elements in the list ✓
b. There should be direct access to the middle element in any sublist
c. The list must be sorted
d. Number values should only be present

Number values should only be present

Question **8**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is small and unsorted ✓
- b. When the list is large and unsorted
- c. When the list is large and sorted
- d. When the list is small and sorted

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Checking each element sequentially ✓
- b. Skipping every second element
- c. Sorting the list before searching
- d. Dividing the list in half repeatedly

Question **10**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(n log n)
- b. O(1)
- c. O(log n) ✓
- d. O👎

Question **11**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. n/2
- b. log n
- c. n ✓
- d. 1

Question **12**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O👎 ✓
- b. O(1)
- c. O(log n)
- d. O(n log n)

Question **13**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

- a. Searching ✓
- b. Mining
- c. Discovering
- d. Finding

Question **14**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Binary search
- b. Linear search ✓
- c. Hash search
- d. Both (1) & (3)

Question **15**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the middle of the list
- b. At the beginning of the list
- c. In the right sublist ✓
- d. In the left sublist

Status	Finished
Started	Friday, 30 October 2026, 3:54 PM
Completed	Friday, 30 October 2026, 4:24 PM
Duration	30 mins 54 secs
Grade	**9.00** out of 10.00 (**90**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Given an list, find peak element in it. A peak element is an element that is greater than its neighbors.

An element a[i] is a peak element if

A[i-1] <= A[i] >=a[i+1] for middle elements. [0<i<n-1]

A[i-1] <= A[i] for last element [i=n-1]

A[i]>=A[i+1] for first element [i=0]

Input Format

The first line contains a single integer n , the length of A .
The second line contains n space-separated integers,A[i].

Output Format

Print peak numbers separated by space.

Sample Input

5

8 9 10 2 6

Sample Output

```
10 6
```

For example:

Input	Result
4 12 3 6 8	12 8

Answer: (penalty regime: 0 %)

```
n = int(input())
a = list(map(int, input().split()))

peaks = []

for i in range(n):
    if i == 0:
        if n == 1 or a[i] >= a[i + 1]:
            peaks.append(a[i])
    elif i == n - 1:
        if a[i] >= a[i - 1]:
            peaks.append(a[i])
    else:
        if a[i] >= a[i - 1] and a[i] >= a[i + 1]:
            peaks.append(a[i])

print(*peaks)
```

	Input	Expected	Got	
✓	7 15 7 10 8 9 4 6	15 10 9 6	15 10 9 6	✓
✓	4 12 3 6 8	12 8	12 8	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Incorrect

Mark 0.00 out of 1.00

String should contain only the words are not palindrome.

Sample Input 1

Malayalam is my mother tongue

Sample Output 1

is my mother tongue

Answer: (penalty regime: 0 %)

```
def remove_palindromes(sentence):
    result = []

    for word in sentence.split():
        cleaned = ''.join(filter(str.isalpha, word)).casefold()

        if cleaned and cleaned != cleaned[::-1]:
            result.append(word)

    return " ".join(result)


sentence = input()
print(remove_palindromes(sentence))
```

	Input	Expected	Got	
✓	Malayalam is my mother tongue	is my mother tongue	is my mother tongue	✓

Your code failed one or more hidden tests.
Your code must pass all tests to earn any marks. Try again.

Incorrect

Marks for this submission: 0.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

An list contains N numbers and you want to determine whether two of the numbers sum to a given number K. For example, if the input is 8, 4, 1, 6 and K is 10, the answer is yes (4 and 6). A number may be used twice.

Input Format

The first line contains a single integer n , the length of list

The second line contains n space-separated integers, list[i].

The third line contains integer k.

Output Format

Print Yes or No.

Sample Input

7

0 1 2 4 6 5 3

1

Sample Output

Yes

For example:

Input	Result
5 8 9 12 15 3 11	Yes
6 2 9 21 32 43 43 1 4	No

Answer: (penalty regime: 0 %)

```
def check_two_sum():
    import sys
    input_data = sys.stdin.read().split()
    if not input_data:
        return

    n = int(input_data[0])

    # Extract the list elements and the target K
    arr = list(map(int, input_data[1:n+1]))
    k = int(input_data[n+1])

    seen = set()
    found = False

    # Iterate through the numbers
    for num in arr:
        # Check if the complement required to reach K exists in our set
        if (k - num) in seen or (num * 2 == k):
            found = True
            break
        seen.add(num)

    # Check if a single number added to itself equals K
    if not found:
        for num in arr:
            if num * 2 == k:
                found = True
                break

```

```
    if found:
        print("Yes")
    else:
        print("No")

if __name__ == "__main__":
    check_two_sum()
```

	Input	Expected	Got	
✓	5 8 9 12 15 3 11	Yes	Yes	✓
✓	6 2 9 21 32 43 43 1 4	No	No	✓
✓	6 13 42 31 4 8 9 17	Yes	Yes	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

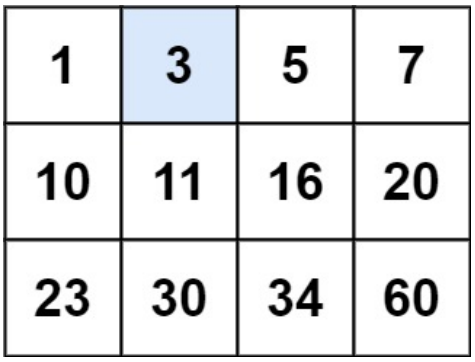
You are given an `m x n` integer matrix `matrix` with the following two properties:

- Each row is sorted in non-decreasing order.
- The first integer of each row is greater than the last integer of the previous row.

Given an integer `target`, return `True` *if* `target` *is in* `matrix` *or* `False` *otherwise*.

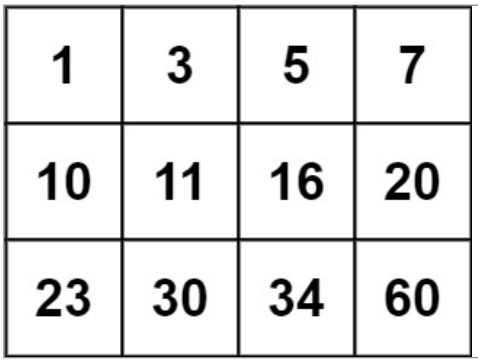
You must write a solution in `O(log(m * n))` time complexity.

Example 1:




```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 3
Output: True
```

Example 2:




```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 13
Output: False
```

For example:

Test	Result
print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13))	False
print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))	True

Answer: (penalty regime: 0 %)

Reset answer

```
def searchMatrix(matrix: list[list[int]], target: int) -> bool:
    if not matrix or not matrix[0]:
        return False

    m = len(matrix)
    n = len(matrix[0])

    left = 0
    right = m * n - 1

    while left <= right:
        mid = (left + right) // 2
        mid_val = matrix[mid // n][mid % n]
```

```
14
15 ▾        if mid_val == target:
16              return True
17 ▾        elif mid_val < target:
18              left = mid + 1
19 ▾        else:
20              right = mid - 1
21
22      return False
```

	Test	Expected	Got	
✓	print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13))	False	False	✓
✓	print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))	True	True	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Given two Strings s1 and s2, remove all the characters from s1 which is present in s2.

Constraints

1<= string length <= 200

Sample Input 1

experience
enc

Sample Output 1

xpri

Answer: (penalty regime: 0 %)

```
def remove_chars(s1: str, s2: str) -> str:
    # Use a set for O(1) character lookup
    chars_to_remove = set(s2)

    # Filter out any character that exists in s2
    result = "".join(char for char in s1 if char not in chars_to_remove)

    return result

# Input reading
s1 = input().strip()
s2 = input().strip()

# Print output
print(remove_chars(s1, s2))
```

	Input	Expected	Got	
✓	experience enc	xpri	xpri	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Given an array `nums` containing `n` distinct numbers in the range `[0, n]`, return *the only number in the range that is missing from the array.*

Example 1:

```
Input: nums = [3,0,1]
Output: 2
Explanation: n = 3 since there are 3 numbers, so all numbers are in the range [0,3]. 2 is the missing number in the range since it does not appear in nums.
```

Example 2:

```
Input: nums = [0,1]
Output: 2
Explanation: n = 2 since there are 2 numbers, so all numbers are in the range [0,2]. 2 is the missing number in the range since it does not appear in nums.
```

Example 3:

```
Input: nums = [9,6,4,2,3,5,7,0,1]
Output: 8
Explanation: n = 9 since there are 9 numbers, so all numbers are in the range [0,9]. 8 is the missing number in the range since it does not appear in nums.
```

For example:

Test	Result
print(missingNumber([3,0,1]))	2
print(missingNumber([0,1]))	2

Answer: (penalty regime: 0 %)

Reset answer

```
def missingNumber(nums: list[int]) -> int:
    n = len(nums)
    return n * (n + 1) // 2 - sum(nums)
```

	Test	Expected	Got	
✓	print(missingNumber([3,0,1]))	2	2	✓
✓	print(missingNumber([0,1]))	2	2	✓
✓	print(missingNumber([9,6,4,2,3,5,7,0,1]))	8	8	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Given an array of integers `nums` which is sorted in ascending order, and an integer `target`, write a function to search `target` in `nums`. If `target` exists, then return its index. Otherwise, return `-1`.

You must write an algorithm with `O(log n)` runtime complexity.

Example 1:

```
Input: nums = [-1,0,3,5,9,12], target = 9
Output: 4
Explanation: 9 exists in nums and its index is 4
```

Example 2:

```
Input: nums = [-1,0,3,5,9,12], target = 2
Output: -1
Explanation: 2 does not exist in nums so return -1
```

Constraints:

- `1 <= nums.length <= 10^4`
- `-10^4 < nums[i], target < 10^4`
- All the integers in `nums` are **unique**.
- `nums` is sorted in ascending order.

For example:

Test	Result
`print(search([-1,0,3,5,9,12],9))`	4

Answer: (penalty regime: 0 %)

Reset answer

```
def search(nums: list[int], target: int) -> int:
    left, right = 0, len(nums) - 1

    while left <= right:
        mid = (left + right) // 2
        if nums[mid] == target:
            return mid
        elif nums[mid] < target:
            left = mid + 1
        else:
            right = mid - 1

    return -1
```

	Test	Expected	Got	
✓	`print(search([-1,0,3,5,9,12],9))`	4	4	✓
✓	`print(search([-1,0,3,5,9,12],2))`	-1	-1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

Two string values S1, S2 are passed as the input. The program must print first N characters present in S1 which are also present in S2.

Input Format:

The first line contains S1.
The second line contains S2.
The third line contains N.

Output Format:

The first line contains the N characters present in S1 which are also present in S2.

Boundary Conditions:

2 <= N <= 10
2 <= Length of S1, S2 <= 1000

Example Input/Output 1:

Input:

abcbde
cdefghbb
3

Output:

bcd

Note:

b occurs twice in common but must be printed only once.

Answer: (penalty regime: 0 %)

```
def print_common_chars():
    s1 = input().strip()
    s2 = input().strip()
    n = int(input().strip())

    set2 = set(s2)
    seen = set()
    result = []

    for char in s1:
        if char in set2 and char not in seen:
            seen.add(char)
            result.append(char)
            if len(result) == n:
                break

    print("".join(result))

if __name__ == "__main__":
    print_common_chars()
```

	Input	Expected	Got	
✓	abcbde cdefghbb 3	bcd	bcd	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Write a Python program for binary search.

For example:

Input	Result
1,2,3,5,8 6	False
3,5,9,45,42 42	True

Answer: (penalty regime: 0 %)

```
def binary_search(arr: list[int], target: int) -> bool:
    left, right = 0, len(arr) - 1

    while left <= right:
        mid = (left + right) // 2
        if arr[mid] == target:
            return True
        elif arr[mid] < target:
            left = mid + 1
        else:
            right = mid - 1

    return False

if __name__ == "__main__":
    # Parsing input list and target integer
    raw_arr = input().strip()
    target = int(input().strip())

    # Convert comma-separated string into a sorted list of integers
    arr = sorted([int(x) for x in raw_arr.split(",") if x.strip()])

    # Print True or False
    print(binary_search(arr, target))

```

	Input	Expected	Got	
✓	1,2,3,5,8 6	False	False	✓
✓	3,5,9,45,42 42	True	True	✓
✓	52,45,89,43,11 11	True	True	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

Balanced strings are those that have an equal quantity of 'L' and 'R' characters.

Given a balanced string s, split it in the maximum amount of balanced strings.

Return the maximum amount of split balanced strings.

Example 1:

Input:

RLRRLLRLRL

Output:

4

Explanation: s can be split into "RL", "RRLL", "RL", "RL", each substring contains same number of 'L' and 'R'.

Example 2:

Input:

RLLLLRRRLR

Output:

3

Explanation: s can be split into "RL", "LLLRRR", "LR", each substring contains same number of 'L' and 'R'.

Example 3:

Input:

LLLLRRRR

Output:

1

Explanation: s can be split into "LLLLRRRR".

Constraints:

1 <= s.length <= 1000

s[i] is either 'L' or 'R'.

s is a balanced string.

For example:

Test	Result
`print(BalancedStrings('RLRRLLRLRL'))`	4
`print(BalancedStrings('RLLLLRRRLR'))`	3

Answer: (penalty regime: 0 %)

Reset answer

```
def BalancedStrings(s: str) -> int:
    balance = 0
    count = 0

    for char in s:
        if char == 'R':
            balance += 1
        elif char == 'L':
            balance -= 1

        if balance == 0:
            count += 1

    return count
```

	Test	Expected	Got	
✓	`print(BalancedStrings('RLRRLLRLRL'))`	4	4	✓
✓	`print(BalancedStrings('RLLLLRRRLR'))`	3	3	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.